

Dear Hiring Committee,

I am excited to apply for the Applications Programmer Associate position within the University of Michigan's Robotics department. As a recent Purdue Computer Science graduate with hands-on experience in full-stack development, I'm drawn to Michigan Engineering's people-first mission and its commitment to building innovative tools that serve the public good.

In my academic and project experience, I've built web applications using JavaScript, Python, and SQL that emphasize usability and maintainability. For example, I developed a Spring Boot and JavaScript-based guitar tablature application that stores user compositions in a MySQL database and uses JWT authentication to ensure secure access. I also created a UX-focused React and Go fretboard diagram builder backed by PostgreSQL, where I designed RESTful APIs and responsive interfaces that are now used by hundreds of musicians. These projects strengthened my skills in clean API design, database integration, and front-end responsiveness, principles I'd bring to developing and supporting software for the Robotics department.

Through my senior capstone, I worked in an Agile/Scrum team to develop a collaborative calendar scheduling system, gaining experience in iterative development, version control, and communication with non-technical stakeholders. I'm eager to bring these skills to the Robotics department, developing reliable applications, improving processes, and supporting your mission of advancing innovation through technology.

Thank you for considering my application. I would be honored to contribute to Michigan Engineering's continued impact and would welcome the opportunity to discuss how my background aligns with your needs.

Sincerely,
Evan Borszem
(317) 474-3734

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EDUCATION

Purdue University

Bachelor of Science in Computer Science

GPA: 3.48 | Dean's List (5 semesters)

Relevant Coursework: Data Structures & Algorithms, Software Engineering, Software Testing, Database Systems, Operating Systems

West Lafayette, IN

May 2025

PROJECTS

Calenduh | *TypeScript, React Native, Go, PostgreSQL, Docker*

- Collaborated on a 5-person Agile team to develop REST APIs for calendar and subscription management, supporting 5,000+ concurrent users
- Optimized PostgreSQL queries and refactored backend routes to improve scalability and response time
- Implemented secure image upload and routing to AWS S3 with strict file validation, enhancing reliability and maintainability

Guitar Diagram Builder | *React, Go, PostgreSQL, CSS, Vercel*

- Launched a responsive web app for creating and exporting guitar diagrams, attracting 250+ users in the first month
- Designed a Go REST API with persistent data storage and a flexible state system supporting custom tunings, scales, colors, and real-time audio
- Added advanced UX features such as SVG/JSON export, dark/light mode, keyboard shortcuts, and left-handed support for accessibility

Guitar Tablature Creator & Player | *Java, Spring Boot, JavaScript, HTML, CSS, MySQL*

- Built a full-stack web app for composing and playing guitar tablatures with real-time playback and interactive fretboard visualization
- Developed REST APIs with JWT-based authentication, implemented MVC architecture, and optimized frontend responsiveness for seamless editing and playback

Shell | *C, Lex, Yacc*

- Created a Unix-like shell supporting I/O redirection, piping, subshell execution, and wildcard expansion using low-level system calls
- Enhanced process management with zombie handling and implemented built-in commands (printenv, setenv, unsetenv, cd, source)

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Go, C, C++, SQL, HTML, CSS

Frameworks/Libraries: Spring Boot, React, JUnit

Tools: Git, Docker, MySQL, PostgreSQL, Maven, Gradle, Postman, AWS S3

Methodologies: Agile, Scrum, REST API design

EXPERIENCE

Target

Small Format Team Member

West Lafayette, IN

June 2024 – May 2025

- Collaborated in a fast-paced, team-oriented environment requiring attention to detail and adaptability
- Developed strong problem-solving and multitasking abilities while managing shifting priorities